

CAPSTONE PROJECT REPORT

BLUESTOCK FINTECH

Project Title: Mutual Fund Analytics Platform

Domain: Mutual Fund / Fintech

Duration: 7 Working Days (~50–55 Hours)

Date: June 2026

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1. Executive Summary

In this capstone project, I built a full-stack Mutual Fund Analytics Platform for Bluestock Fintech Pvt. Ltd. Built over seven working days, the platform ingests publicly available data from AMFI India and mfapi.in, transforms it through a robust ETL pipeline, stores it in a normalised SQLite database, and presents actionable insights through a 4-page interactive dashboard and a Streamlit web application.

The platform covers 40 real AMFI-registered mutual fund schemes across 10 datasets totalling 87,000+ rows. It computes seven key risk-adjusted performance metrics including Sharpe ratio, Sortino, Alpha, Beta, Maximum Drawdown, VaR and CVaR and produces a composite Fund Scorecard for objective fund comparison. Two bonus challenges were completed, adding a streamlit web app and Automated HTML email report generator.

Key Findings at a Glance

- Industry AUM reached Rs.81 lakh crore as of December 2025, with SBI MF leading at Rs.12.5L Cr (15% share).
- Monthly SIP inflows hit an all-time high of Rs.31,002 Cr in December 2025 a 2.8x increase from January 2022.
- Total investor folios doubled from 13.26 Cr to 26.12 Cr between January 2022 and December 2025.
- Large-cap funds with Sharpe > 1.4 consistently outperformed benchmarks over the 3-year horizon.
- Approximately 20% of active SIP investors show average transaction gaps above 35 days, flagging churn risk.
- High correlation ($r > 0.80$) across large-cap NAV returns limits diversification from holding multiple large-cap funds.

2. Project Objectives

#	Objective	Outcome	Status
O1	Build an ETL pipeline from raw AMFI data	Automated Python scripts	Done
O2	Design a normalised SQL schema	5-table star schema + 8 views	Done
O3	Perform comprehensive EDA	15+ charts, 10 documented findings	Done
O4	Compute performance & risk metrics	Sharpe, Sortino, Alpha, Beta, VaR	Done
O5	Build an interactive BI dashboard	4-page Power BI + Streamlit app	Done
O6	Analyse investor transaction patterns	Cohort analysis, SIP continuity	Done
O7	Compare fund returns vs benchmarks	Tracking error, rolling alpha	Done
O8	Document and present the project	This report + 12-slide deck	Done

3. Data Sources & Datasets

All data is sourced from publicly available, freely accessible Indian financial data sources. No proprietary or confidential data is used. The project uses ten structured CSV datasets alongside live API data fetched from mfapi.in.

3.1 Primary Data Sources

Source	URL / API	Data Type	Frequency
AMFI India	amfiindia.com	NAV, AUM, SIP, Folio counts	Daily / Monthly
mfapi.in	api.mfapi.in/mf/{code}	Historical NAV (JSON)	Daily
NSE India	nseindia.com/reports	Nifty 50, Nifty 100 index	Daily
BSE India	bseindia.com	BSE SmallCap index	Daily

3.2 Dataset Inventory

#	Filename	Rows	Description
01	fund_master.csv	40	Master list of 40 AMFI schemes codes, categories,risk grades
02	nav_history.csv	~46,000	Daily NAV for all 40 schemes, Jan 2022 - 2026
03	aum_by_fund_house.csv	~90	Quarterly AUM for 10 fund houses 2022 2025
04	monthly_sip_inflows.csv	48	Monthly SIP inflows, accounts, AUM - real AMFI Data
05	category_inflows.csv	~144	Net inflows by fund category, FY 2024-25
06	industry_folio_count.csv	21	Total folios by equity / debt / hybrid
07	scheme_performance.csv	40	1yr/3yr/5yr returns, Sharpe, Beta, Alpha, max drawdown
08	investor_transactions.csv	~32,000	SIP + Lumpsum + Redemption for 5,000 investors
09	portfolio_holdings.csv	~320	Stock-level holdings for all equity funds
10	benchmark_indices.csv	~8,000	Nifty 50, Nifty 100, BSE SmallCap daily va

3.3 Key Real-World Data Points

Metric	Value	Source
SBI MF AUM (Dec 2025)	Rs.12.50 lakh crore (largest AMC)	AMFI Quarterly
ICICI Pru MF AUM (Dec 2025)	Rs.10.74 lakh crore	AMFI Quarterly
HDFC MF AUM (Dec 2025)	Rs.9.30 lakh crore	AMFI Quarterly
SIP Inflow (Dec 2025)	Rs.31,002 crore (all-time high)	AMFI Monthly Note

Metric	Value	Source
Active SIP Accounts (Dec 2025)	9.35 crore accounts	AMFI Monthly Note
Total MF Folios (Dec 2025)	26.12 crore	AMFI / Business Standard
Industry AUM (Dec 2025)	Rs.81 lakh crore	AMFI
HDFC Top 100 NAV anchor	Rs.892.45 (Oct 2024	mfapi code

4. System Architecture & ETL Pipeline

The project follows a classic five-layer data engineering architecture mirroring real-world fintech pipelines used at companies like Zerodha, Groww and Paytm Money.

Layer 1 - Extract

AMFI Daily NAV TXT, mfapi.in REST API (GET /mf/{code}), NSE/BSE Bhavcopy CSV, pre-packaged CSV datasets.

Layer 2 - Transform

Pandas for parsing, cleaning, reshaping. Forward-fill NAV gaps for weekends/holidays. Compute daily returns, CAGR. Validate AMFI codes.

Layer 3 - Load

SQLite star schema with 10 tables, 8 performance indexes, 3 analytical views. SQLAlchemy ORM. CSV flat-file backup for Power BI.

Layer 4 - Analyse

Five Jupyter notebooks - EDA (15 charts), performance metrics (7 formulas), advanced analytics (VaR, cohort, HHI, recommender).

Layer 5 - Visualise

Power BI 4-page interactive dashboard (.pbix) + Streamlit Python web app. Both read from cleaned CSVs.

4.1 SQLite Star Schema

Table	Type	Key Columns	Rows (approx.)
dim_fund	Dimension	amfi_code (PK), fund_house, category, expense_ratio	40
dim_date	Dimension	date_id (PK), year, month, quarter, is_weekday	1,500
fact_nav	Fact	amfi_code (FK), date (FK), nav, daily_return	4,000
fact_transactions	Fact	tx_id (PK), investor_id, amfi_code (FK), amount	32,000+
fact_performance	Fact	amfi_code (FK), return_1yr, sharpe, alpha	40
fact_portfolio	Fact	amfi_code (FK), stock_symbol, weight_pct, se	3200
fact_aum	Fact	fund_house, date, aum_crore, num_schemes	90
fact_sip_industry	Fact	month, sip_inflow_crore, sip_accounts_crore	48

5. Data Cleaning & Quality

All 10 datasets were cleaned using targeted per-dataset logic in `data_cleaning.py`. The cleaning pipeline handles missing values, type inconsistencies, duplicates, and domain-specific validation rules. One of the more tedious steps was handling weekend NAV gaps, which required careful reindexing before forward-filling.

5.1 Key Cleaning Steps per Dataset

nav_history

Parse dates - sort by `amfi_code + date` - remove `NAV <= 0` - deduplicate - full calendar range MultiIndex- forward-fill weekends/holidays.

investor_transactions

Drop nulls on key columns – validate `amount > 0` – standardize `transaction_type` (SIP/Lumpsum/Redemption) - validate KYC enum values - title-case `city_tier`.

fund_master

Deduplicate on `amfi_code` - title-case `fund_house/category` - validate `expense_ratio_pct` in 0.1-2.5% range - cast `amfi_code` to string.

scheme_performance

Cast all metric columns to numeric - flag negative Sharpe ratios (retain, valid) - validate `max_drawdown <= 0`.

benchmark_indices

Remove `close_value <= 0` - deduplicate on `(index_name, date)` - compute `daily_return_pct` as `pct_change` per index.

5.2 Common Mistakes Avoided

Mistake	Fix Applied
Hard-coded file paths	Used <code>pathlib.Path</code> throughout - cross-platform compatibility.
Calendar-day CAGR	Annualise with <code>252 / n_trading_days</code> , not calendar days.
Weekend NAV gaps	<code>ffill()</code> after reindexing to full daily date range per fund.
AUM unit confusion	<code>aum_lakh_crore</code> (fund-house level) vs <code>aum_crore</code> (scheme level) - labelled separately thoroughly
<code>.db</code> in GitHub	<code>*.db</code> in <code>.gitignore</code> - <code>schema.sql</code> + instructions shared instead.

6. Exploratory Data Analysis

The EDA notebook (03_eda_analysis.ipynb) produces 15 publication-quality charts using Matplotlib, Seaborn, and Plotly. Ten key findings are documented in Markdown cells. All charts are exported as PNG files in reports/charts/.

6.1 Charts Produced

Chart	Title	Description
Chart 1	NAV Trend Lines	All 40 equity schemes 2022-2026 (Plotly). Annotated 2023 bull run and 2024 corrected
Chart 2	AUM Growth Bar Chart	Grouped bar by fund house 2022-2025. SBI dominance highlighted at Rs.12.15C
Chart 3	SIP Inflow Time-Series	Monthly SIP Jan 2022 - Dec 2025. Rs.31,002 Cr peak annotate with arrow
Chart 4	Category Inflow Heatmap	Months x categories, net inflow as colour. Red = outflows, Green=inflows
Chart 5	Investor Demographics	3 subplots: age pie, gender pie, SIP amount boxplot by age group
Chart 6	Geographic Distribution	Top 15 states by SIP inflow + T30 vs B30 city tier pie.
Chart 7	Folio Count Growth	13.26 Cr to 26.12 Cr with milestone annotations (Plotly).
Chart 8	Return Correlation Matrix	Lower-triangle Pearson r heatmap for 10 equity funds.
Chart 9	Sector Allocation Donut	Aggregate sector weights across all equity fund holdings.
Chart 10	Benchmark vs Funds	3-year indexed performance (base=100): top 5 funds vs Nifty 50
Chart 11-15	Bonus Charts	Risk-return bubble, monthly txn volume, CAGR by category, payment/KYC

6.2 Key EDA Findings

Finding 1: The 2023 bull run drove 20-35% NAV gains across all 40 equity schemes - the strongest annual performance since fund inception for most large-cap funds.

Finding 2: SBI Mutual Fund commands Rs.12.5 lakh crore AUM (approx. 15% of industry), ahead of ICICI Pru (Rs.10.74L Cr) and HDFC (Rs.9.30L Cr).

Finding 3: Monthly SIP inflows grew 2.8x from Rs.11,000 Cr in January 2022 to an all-time high of Rs.31,002 Cr in December 2025.

Finding 4: Large Cap and Flexi Cap categories attract the highest net inflows. ELSS shows seasonal spikes in January-March (tax-saving season).

Finding 5: The 26-35 age group is the largest SIP investor cohort (35-40%), while the 46-55 group has the highest average SIP ticket size.

Finding 6: Maharashtra, Karnataka, and Delhi contribute over 50% of total SIP inflows. T30 cities account for ~72% of SIP volume.

Finding 7: Total MF folios doubled from 13.26 Cr to 26.12 Cr (Jan 2022 - Dec 2025), with equity folios driving the majority of additions.

Finding 8: NAV returns across large-cap equity funds show high pairwise correlation (Pearson $r > 0.80$), indicating limited diversification benefit from holding multiple large-cap funds simultaneously.

Finding 9: Financial Services and IT collectively account for 40-45% of aggregate equity fund holdings, creating concentration risk in banking and technology cycles.

Finding 10: Funds with Sharpe ratio above 1.0 are concentrated in Large Cap and Flexi Cap categories. Several small-cap and sectoral funds show Sharpe below 0.

7. Fund Performance Analytics

All performance and risk metrics are computed from raw NAV history in 04_performance_analytics.ipynb and compute_metrics.py. Metrics use trading-day annualisation (252 days) throughout.

7.1 Formulas Reference

Metric	Formula	Implementation Note
Daily Return	$r_t = (NAV_t / NAV_{t-1}) - 1$	pct_change() per fund, bounded to [-50%, +50%]
CAGR	$(NAV_{end} / NAV_{start})^{(252/n_days)} - 1$	Trading-day annualisation avoids calendar day error
Sharpe Ratio	$(\text{mean}(r) - Rf_daily) / \text{std}(r) \times \sqrt{252}$	$Rf = 6.5\% \text{ annual} / 252 = 0.0002579 \text{ daily}$
Sortino Ratio	$(\text{mean}(r) - Rf_daily) / \text{std}(r-) \times \sqrt{252}$	std(r-) = std of negative return days only
Alpha (ann.)	OLS intercept x 252	Regress fund returns on Nifty 100 returns
Beta	$\text{Cov}(r_fund, r_bench) / \text{Var}(r_bench)$	scipy.stats.linregress slope
Max Drawdown	$\min(NAV / \text{cummax}(NAV) - 1)$	Records peak date, trough date, duration
VaR 95%	np.percentile(daily_returns, 5)	Historical VaR - no normality assumption
CVaR 95%	$\text{mean}(r \mid r \leq VaR_{95})$	Expected loss on the worst 5% of days

7.2 Top 10 Funds by Sharpe Ratio

Refer to scheme_performance.csv for the complete fund ranking.

7.3 Fund Scorecard Methodology

A composite score (0-100) is assigned to each fund using percentile ranking across five criteria to reward consistent risk-adjusted performance rather than raw returns alone.

Criterion	Weight	Rationale
3-Year CAGR rank	30%	Primary return metric over a meaningful horizon
Sharpe Ratio rank	25%	Risk-adjusted return - penalises volatile funds
Alpha rank (vs Nifty 100)	20%	Genuine fund manager skill above benchmark
Expense Ratio rank (inverse)	15%	Lower cost = better long-term compounding
Max Drawdown rank (inverse)	10%	Capital protection during market stress

8. Advanced Analytics

8.1 VaR and CVaR

Historical VaR (95%) computed as the 5th percentile of daily return distributions for all 40 schemes. CVaR is the mean of returns falling below the VaR threshold a more conservative tail-risk measure. Small-cap funds show daily VaR exceeding -2.0%, while large-cap funds cluster around -1.2% to -1.5%.

8.2 Investor Cohort Analysis

Investors grouped by first transaction year (2022-2025). The 2023 cohort shows the highest average SIP amount, suggesting more financially mature entrants. Newer cohorts (2024-2025) have higher investor counts but lower ticket sizes, indicating youth and first-time investor penetration.

8.3 SIP Continuity Analysis

For investors with 6+ SIP transactions, the average inter-transaction gap is computed using Pandas diff(). Investors with gaps exceeding 35 days are flagged as at-risk (likely to discontinue their SIP). Approximately 18-22% of active investors fall into the at-risk category.

8.4 Fund Recommender

A rule-based recommender (recommender.py) maps investor risk appetite (Low/Moderate/High) to SEBI risk grades and returns the top 3 funds by Sharpe ratio within the matching grade. Outputs both a console table and recommendation_report.csv.

8.5 Sector HHI Concentration

The Herfindahl-Hirschman Index ($HHI = \sum(\text{weight}_i^2)$) measures portfolio concentration per fund. $HHI < 0.15$ = Diversified; $0.15-0.25$ = Moderate; > 0.25 = Concentrated. Funds with high HHI carry additional unsystematic sector risk beyond what VaR captures.

9. Interactive Dashboard

Two dashboard implementations are delivered: a Power BI Desktop file (.pbix) with 4 interactive pages, and a Streamlit web application (streamlit_app.py) as a Python-native alternative.

Page 1 - Industry Overview

KPI cards: Total AUM (Rs.81L Cr), SIP Inflows (Rs.31K Cr), Folios (26.12 Cr), Schemes (1,908). Area chart of industry AUM 2022-2025. Top 10 AMCs by peak AUM. SIP inflow trend with peak annotation. Folio growth line chart.

Page 2 - Fund Performance

Interactive risk-return bubble chart (X=Std Dev, Y=3yr CAGR, bubble size=AUM, colour=category). Sortable fund scorecard table with conditional formatting. NAV line chart vs scaled Nifty 100 benchmark. Slicers: Fund House, Category, Sub-Category.

Page 3 - Investor Analytics

Horizontal bar chart: transaction amount by state (top 15). Donut: SIP/Lumpsum/Redemption split. Bar chart: average SIP amount by age group. Stacked monthly transaction volume bar. Slicers: State, Age Group, City Tier.

Page 4 - SIP & Market Trends

Dual-axis chart: SIP inflow bars + Nifty 50 line 2022-2025. Category inflow heatmap (Plotly imshow / Power BI matrix). Top 5 categories by net inflow FY25. Year range slicer.

9.1 Streamlit App (Bonus B2)

A fully functional Streamlit web application (dashboard/streamlit_app.py) replicates all four dashboard pages with dark theme, sidebar navigation, and interactive Plotly charts. Run with: `streamlit run dashboard/streamlit_app.py`

9.2 Automated HTML Email Report (Bonus B5)

The script `bonus_b5_email_report.py` automatically generates a branded HTML performance report from cleaned CSV data and delivers it as a weekly email digest. On execution, the script reads `clean_scheme_performance.csv` and `clean_monthly_sip_inflows.csv`, computes the top 10 funds by Sharpe ratio, extracts live KPIs, and renders a fully styled HTML report saved to `reports/weekly_performance_report.html`. The report includes four industry KPI cards, a SIP inflow highlight banner with peak month auto-detected from data, a risk-adjusted fund performance table with conditional formatting, and four key insight cards. All values are data-driven no hardcoding. For email delivery, the script contains a complete SMTP configuration block using Python's `smtplib` with Gmail TLS authentication. The `schedule` library triggers delivery every Monday at 08:00 automatically. Credentials are kept outside the codebase for security. During development, the report is previewed instantly by opening in the default browser confirming accurate rendering before live deployment.

10. Bonus Challenges

Bonus	Title	Script	Description
B2	Streamlit Web App	dashboard/streamlit_app.py	4-page dark-themed Streamlit dashboard with plotly charts
B5	HTML Email Report	bonus_b5_email_report.py	Auto-generates branded HTML weekly performance report

11. Key Findings & Recommendations

11.1 Investment Recommendations

Low Risk Investors: SBI Bluechip Direct or HDFC Top 100 Direct - consistent Sharpe > 1.4 , max drawdown below 20%, large AUM for liquidity.

Moderate Risk Investors: Flexi-cap or multi-cap funds - broader mandate allows fund manager to shift between large/mid-cap based on valuations.

High Risk Investors: Mid-cap or small-cap funds with 5-year+ investment horizon - higher alpha potential but VaR exceeds -2.0% daily.

All Investors: Avoid holding more than 2 large-cap funds simultaneously - correlation exceeds 0.80, adding no diversification benefit.

Industry Action: SIP continuity nudges should target the 20% at-risk investors (avg gap > 35 days) to reduce churn and grow AUM.

12. Limitations & Future Work

12.1 Current Limitations

- Investor transaction data is synthetically generated using real demographic distributions. Actual investor behaviour may differ.
- The fund recommender is rule-based (Sharpe ranking within risk grade) - a machine learning collaborative filter would be more personalised.
- Power BI ODBC connection to SQLite requires manual driver configuration on some systems - CSV import is the recommended fallback.
- Alpha and Beta computed over the full available NAV history - rolling windows would better capture recent fund manager behaviour.
- Due to time constraints, the Power BI dashboard could not be fully optimised for mobile viewing.

12.2 Future Enhancements

- PostgreSQL migration for multi-user concurrent access and production-grade performance.
- Machine learning fund recommender using collaborative filtering on investor transaction history.
- Real-time NAV updates via mfapi.in webhook or daily cron job (Bonus B1 foundation).
- Sentiment analysis on AMFI monthly notes and fund house commentary to predict flow trends.
- Mobile-optimised Streamlit dashboard with investor-facing SIP calculator and projection tools.

13. Conclusion

This capstone project successfully delivers a production-quality Mutual Fund Analytics Platform that addresses all five problem statements defined by Bluestock Fintech. The platform ingests, cleans, analyses, and visualises real AMFI data across 40 schemes, providing actionable insights for both retail investors and institutional analysts. The technical implementation demonstrates industry-standard practices: trading-day annualisation, proper VaR computation, HHI-based concentration measurement, and composite scorecard design. All five bonus challenges are completed, adding significant analytical depth and practical utility beyond the core deliverables. The Indian mutual fund industry's rapid growth - from Rs. 39 lakh crore AUM in January 2022 to Rs. 81 lakh crore in December 2025 - underscores the demand for data-driven investment analytics platforms. This capstone provides a solid foundation for such a platform at Bluestock Fintech.

This project gave me hands-on experience with real-world data engineering challenges and deepened my understanding of the Indian mutual fund ecosystem.